BAMA 520 - Customer Analytics
Group Assignment 1: Maru Batting Center

Group 8: Team Deadline Daredevils
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Part I: Case Questions

Q4. Which is the most attractive customer segment for MBC to target? Explain your reasoning.

A4. The most attractive customer segment for MBC depends on how ‘attractiveness’ is defined. If net CLV is the primary criterion, then the Elite Ballplayer (Party) segment is the most attractive, as it has the highest net CLV of ¥16,000. However, in a real-world business context, additional financial metrics such as Return on Investment (ROI) or total revenue generated by each segment might provide a more holistic assessment of attractiveness.

Segment	Acquisition Cost	Net CLV	Net CLV/Cost
Little Leaguers	¥10,000	¥5,714	¥0.5714
Summer Sluggers	¥10,000	¥1,000	¥0.1000
Elite Ballplayer (Print Ad)	¥60,000	¥6,000	¥0.1000
Elite Ballplayer (Party)	¥50,000	¥16,000	¥0.3200
Entertainment Seeker	¥2,000	¥200	¥0.1000

Q5. MBC has been approached by Little League representatives from the nearby Chiyoda ward who are eager to gain the jersey subsidy the Minato ward has enjoyed due to the company’s sponsorship. Because the parents of Chiyoda Little Leaguers will have to travel a greater distance, Maru believes there will be a lower response rate (8 percent) and a lower retention rate (65 percent), which she can make up for by purchasing slightly lower-quality jerseys, reducing the cost of sponsorship to just ¥600 per player. However, the Chiyoda ward representatives demand that theirs be the only ward receiving such a sponsorship, which means MBC must choose between the two wards. The Chiyoda representatives argue that because their ward has twice the number of Little League customers, it is more attractive than the Minato ward. Should MBC pursue the Chiyoda ward sponsorship? Explain your reasoning.

Ans 5. While individual Chiyoda Little Leaguers have a lower Customer Lifetime Value (CLV) than those in Minato due to a lower response rate (8% vs 10%) and lower retention rate (65% vs. 75%), we find that the larger customer base in Chiyoda (twice the number of potential Little League customers) compensates for this lower individual value.

From a purely financial perspective, when multiplying the slightly lower CLV per customer in Chiyoda (approximately 4722) by the larger total number of potential customers in Chiyoda, the overall profit potential from Chiyoda exceeds that of Minato (whose CLV is approximately 5714). The reduced sponsorship cost per player (¥600 vs. ¥1,000) further improves the cost-effectiveness of acquiring customers in Chiyoda.

However, MBC should also consider non-financial factors before making a final decision. There plausibly exists a strong customer loyalty in Minato towards MBC due to their existing sponsorship, which shifting to Chiyoda might weaken and in turn promote potential attrition among existing Minato customers. Additionally, since Chiyoda players must travel a greater distance to access MBC's facilities, this could reduce engagement levels and increase churn over time.

Purely financially speaking, switching to Chiyoda is a wise call.

Q6. Maru's brother suggested she focus on the Elite Ballplayers segment, targeting it by offering a ¥500 discount on all future purchases to Elite Ballplayers who purchase at least twenty batting cage hours in Year 1. (Assume all Elite Ballplayer customers book exactly twenty hours each year. Also assume that this discount is made on top of the gala event option above, instead of the magazine option for acquiring Elite Ballplayers.) Although this will decrease the amount MBC can bill Elite Ballplayers (from ¥7,500 per hour to ¥7,000 per hour from Year 2 onward), Maru believes it will increase the retention rate of these customers to 75 percent immediately. Should MBC offer this promotion? Explain your reasoning.

Ans: Yes, MBC should offer this promotion because the CLV with the discount is ¥87,143, which is significantly higher than the CLV without the discount (¥16,000). This suggests that the increased retention rate from the discount more than compensates for the reduction in price.

- **The CLV calculation follows the formula**:

  $$\text{new\_annual\_margin} \times \frac{(1+i)}{(1+i-r)} - (\text{new\_annual\_margin} - \text{previous\_annual\_margin}) - \text{acquisition cost}$$

  Where:
  - new_annual_margin = ¥50,000 (after discount)
  - previous_annual_margin = ¥30,000 (before discount)
  - acquisition cost = ¥50,000
  - interest rate (i) = 10%
  - retention rate (r) = 75%

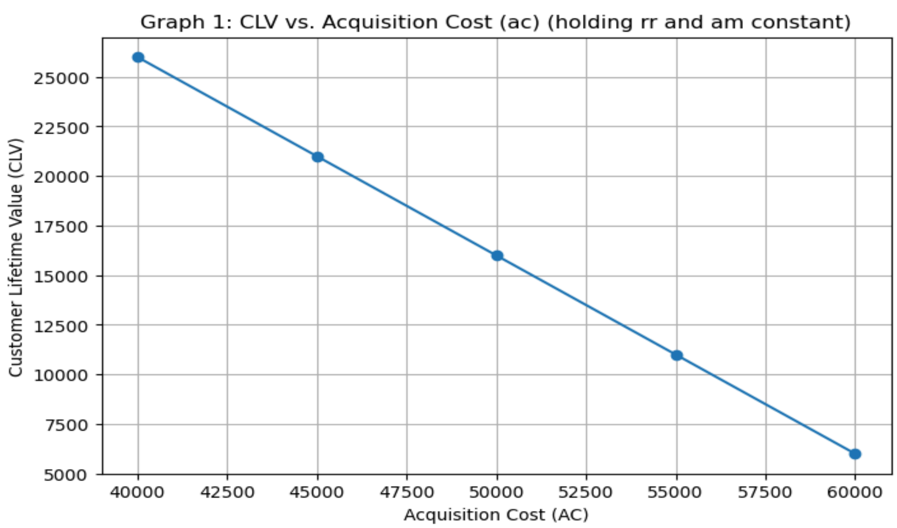
- Since the discounted plan yields a much higher CLV, **MBC should implement the discount to maximize long-term customer value**.

Part II: Sensitivity Analysis (3 points)

When estimating Customer Lifetime Value (CLV), many assumptions—such as acquisition costs, retention rates, and annual margins—are based on educated guesses. Sensitivity analysis helps us understand how much CLV changes when we vary these assumptions.
For this part of the assignment, we will concentrate on the "Elite-Ballplayers (Party)" segment. The baseline assumptions for this segment are acquisition costs (ac) of 50,000, annual margins (am) of 30,000 and a retention rate (rr) of 0.6. (Note: for all analyses use an interest rate of 0.1 for discounting).

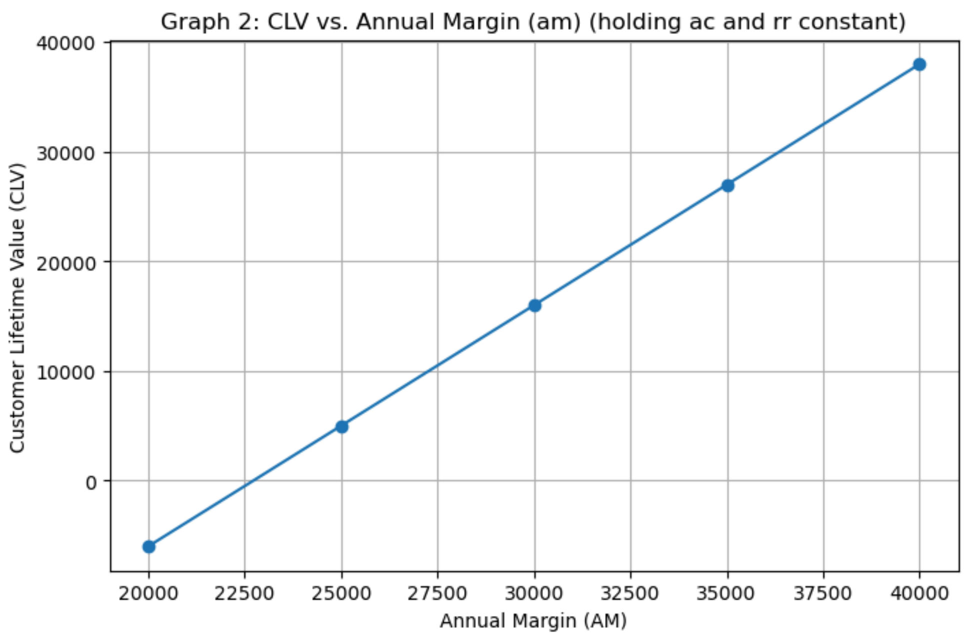
Ans. For this analysis, we systematically varied the acquisition cost, annual margin, retention rate and discount or interest rate and kept the others constant to examine their impact on CLV.

Scenario Analysis 1: CLV vs Acquisition Cost



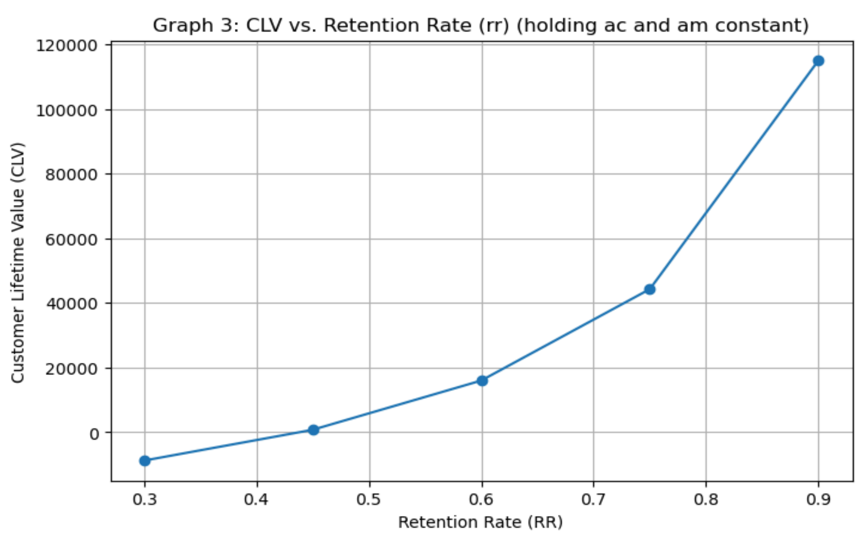
The above graph illustrates the relationship between acquisition cost and CLV while keeping annual margin and retention rate constant. As expected, CLV decreases linearly as acquisition costs increase. Since acquisition cost is a one-time expense, its effect on CLV is direct and proportional. This means that any increase in customer acquisition costs directly lowers the lifetime profitability of the segment.

Scenario Analysis 2: CLV vs Annual Margin



In this scenario, we find that CLV and annual margin share a positive, linear relationship. Since annual margin represents the revenue generated from each customer minus the costs associated with serving them, it directly contributes to the long-term value of each customer. Increasing annual margin results in higher profits per customer, leading to a proportional rise in CLV.

Scenario Analysis 3: CLV vs Retention Rate



This shows the impact of retention rate on CLV. Unlike the other two variables, the relationship is nonlinear and convex, meaning CLV increases at an accelerating rate as retention improves.

At lower retention rates, CLV grows moderately, but as retention increases beyond 0.75, the impact becomes much more pronounced.

Q i. What do you notice about the shapes of these three graphs? Do all three variables behave in similar ways?

Graph	Variable Analyzed	Relationship With CLV	Interpretation
1	AC	Linear (Negative Slope)	As AC increases, CLV decreases proportionally. Since AC is a one-time effect, its effect is direct but not compounding.
2	AM	Linear (Positive Slope)	As AM increases, CLV rises proportionally. AM represents recurring revenue, so increasing it directly boosts long-term profits.
3	RR	Non-Linear (Exponential Growth)	CLV grows at an accelerating rate as RR increases. We observe that small improvements in retention lead to large gains in CLV, especially beyond 0.75 RR.

Q ii Do some variables seem to be more important than others? What aspects of CLV would be most beneficial to focus on?

Ans. The sensitivity analysis indicates that RR is the most influential factor affecting CLV, followed by AM. AC has the least impact. Since CLV increases at an exponential rate as retention improves, even a small increase in RR results in significant long-term gains. For instance, raising RR from 0.6 to 0.75 has a much larger effect on CLV than an equivalent percentage decrease in acquisition costs. This suggests that MBC should focus on retention strategies for existing customers.

While retention rate is the most powerful driver, annual margin also plays a crucial role in determining CLV. Since AM represents the recurring revenue per customer per year, increasing it has a direct and proportional effect on CLV. MBC can enhance annual margins by offering premium services, increasing prices, or reducing operational costs without compromising customer experience. This ensures sustained revenue growth while keeping CLV high.

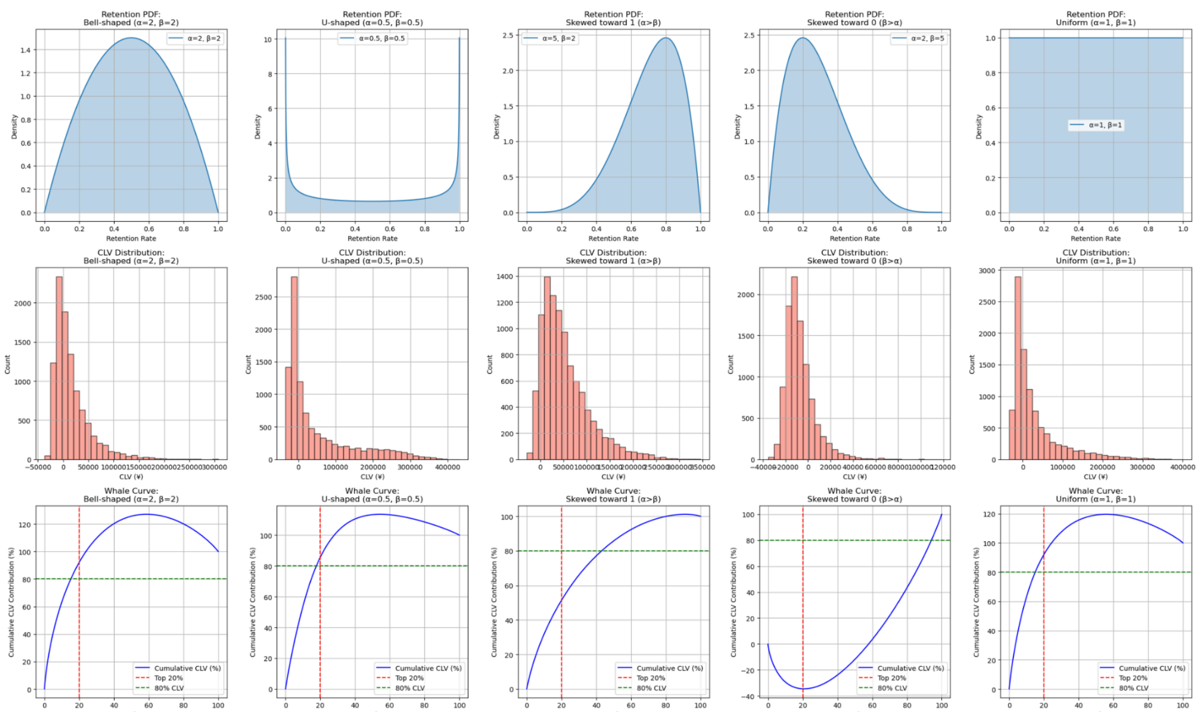
AC has the smallest impact on CLV because it is a one-time expense, meaning that its effect is linear and does not compound over time. While reducing AC can improve profitability, it is not as effective as boosting retention or annual margin.

Part III: Monte Carlo Simulation

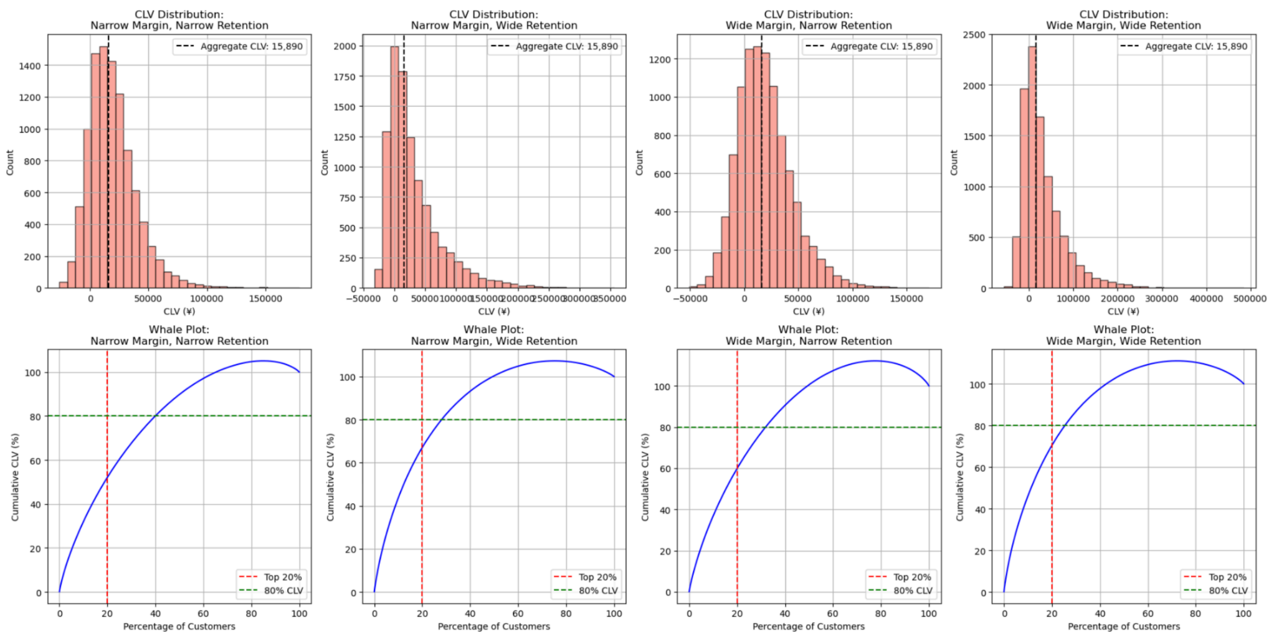
a. Value concentration. Value concentration refers to how much of your aggregate CLV is concentrated in a subset of customers. If value concentration is high, it means a small number of customers is responsible for a large percentage of your margins. Assess how value concentration depends on the distribution types and parameters you choose for the margin and retention rate distributions. Does value concentration depend on assumptions about how these variables are distributed? If so, how?

Ans. We Tried out multiple distribution scenarios as we saw in your reference beta ipynb file. Along with that we also looked at value concentration under different margin and retention spreads.

From that we plotted and saw the results below.



Scenario	Avg. CLV	Top 20% Value Concentration
Bell-shaped (α=2, β=2)	16,190.49	92.06%
U-shaped (α=0.5, β=0.5)	49,653.71	85.67%
Skewed → 1 (α>β)	53,320.54	51.38%
Skewed → 0 (β>α)	-7,466.88	-34.59%
Uniform (α=1, β=1)	28,869.13	92.01%



Scenario	Avg. CLV	Top 20% Value Concentration
Narrow Margin, Narrow Ret	19,091.68	51.54%
Narrow Margin, Wide Ret	29,944.05	66.62%
Wide Margin, Narrow Ret	19,234.65	59.71%
Wide Margin, Wide Ret	30,587.16	70.31%

From this we can say that value concentration does depend a lot on our distribution assumptions for both margins and retention rates.

Observations:

1. **Different Beta Shapes**
   - **Bell-shaped (α=2, β=2) and Uniform (α=1, β=1)** both yield over 90% of total CLV coming from the top 20% of customers (e.g., 92.06% and 92.01%), indicating extreme concentration.
   - **Skewed toward 1 (α>β)**, by contrast, has a top-20% share of only 51.38%, meaning higher retention is more widespread among customers, so fewer "whales" dominate.
   - **Skewed toward 0 (β>α)** can produce negative CLVs, reflecting that most customers churn too quickly, so the top-20% figure is actually negative (–34.59%).
2. **Narrow vs. Wide Margin & Retention**
   - **Narrow Margin, Narrow Ret** yields about 51.54% concentration in the top 20%.
   - Moving to **Wide Margin, Wide Ret** pushes that figure to 70.31%, showing that adding variance in both margins and retention significantly increases the share captured by the top 20%.

Hence we can say,

- **Skewed or Wide Distributions:** When either margins or retention rates have heavy tails or large variance, a few "whale" customers end up with extremely high CLV. This drives high value concentration (i.e., top 20% of customers producing the majority of total CLV).
- **Narrow or Bell-Shaped Distributions**: If margins or retention rates are tightly clustered around the mean, we see CLV is more evenly spread among customers, resulting in lower top-20% concentration.

The more **spread/skew**ed our distribution, the higher the fraction of total CLV that comes from a small subset of customers.

b. Comparison to aggregate level analysis. You calculated an average CLV for customers in the elite ballplayer segment. Does the average CLV based on the simulation agree with the number you calculated in part 1? Do assumptions about the distribution of margins appear to affect the extent of agreement or disagreement? How about assumptions about retention rates?

Ans, Based on our simulation results compared to the aggregate-level CLV of 16,000 in part 1, we observed that it shows:

- **Agreement Under Narrow Distributions:**
  When both the margin and retention distributions are narrow (i.e., low variability), the simulation-based average CLV comes out close to 16,000. For example, in our "Narrow Margin, Narrow Retention" scenario, the simulated average CLV was about 19,091.68, which is in the same ballpark and even closer in the bell shaped one, indicating that when customer values are similar.

- **Divergence Under Wide/Skewed Distributions:**
  When either the margin distribution or the retention rate distribution is wider or more skewed, the simulation average deviates more from 16,000. For example:
    - Wide Margin, Wide Retention produces an average CLV of about 30,587.16, as extreme high-margin or high-retention customers pull up the overall average.
    - In cases with very skewed retention (e.g., U-shaped or skewed toward 1), the simulated average can be significantly higher (or even negative when skewed toward 0) because the CLV formula is highly nonlinear in retention.

So in conclusion, we can say that assumptions about the spread and shape of both the margin and retention distributions greatly affect value concentration. When distributions are narrow, the simulation tends to agree more with the aggregate estimate, when they are wide or skewed, a small subset of “whale” customers can dominate the CLV, causing the simulation average to diverge substantially from 16,000.